AUSTRALIAN INSTITUTE
OF HIGHER EDUCATION

Unit Code: MBIS4017

Unit Name: Cybersecurity Risk Management and Strategies

Study Period: B6 **Year** 2024

Unit Outline

Subject Information

Term Dates	Start: 9/9/2024 End: 4/10/2024
Session Times	Monday/Wednesday: 1pm-5pm Syd/Mel
Awards	Master of Business Information Systems (Cybersecurity)
Prerequisites/ Corequisites	None
Delivery Method	Lecture/Tutorial (on campus)
Unit Description	MBIS4017 Cyber Security Risk Management introduces students to the background of cyber security and cyber-risk management. It covers a range of risk management techniques for cybersecurity based on established international standards (including ISO 31000) and industrial best practices. Topics include information security, cyber security, cyber-risk management, cyber risk assessment (risk identification, analysis, evaluation and treatment) and how to address risk measurement, risk scales, uncertainty and high-consequence risk with low likelihood ¬ in practice, as well as business continuity management. The use of AI should only be undertaken if an assessment clearly states that it is permitted. Penalties will apply if AI is used intentionally without permission.

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Subject Information (cont'd)

Learning Outcomes and Graduate Attributes	On successful completion, students will have achieved the following learning outcomes:	
	Unit Learning Outcomes	**AIH Graduate Attributes**
	ULO01: Discuss existing cyber security risk pertaining to information systems, including common vulnerabilities, potential threats, and possible repercussions of a security breach.	1. Communication 3. Research 4. Critical thinking & problem solving
	ULO02: Relate identified cyber-risks in information systems to the more commonly occurring risks associated with business and/or project management.	4. Critical thinking & problem solving
	ULO03: Analyse and explain known challenges in cyber security practices in relation to risk assessments.	1. Communication 3. Research 4. Critical thinking & problem solving
	ULO04: Develop organisational strategies and recommendation for cyber-risk management.	1. Communication 2. Collaboration 4. Critical thinking & problem solving
	ULO05: Prepare and communicate cyber security risk management and business continuity plan.	1. Communication 2. Collaboration 4. Critical thinking & problem solving

Contact Details	**Unit Co-ordinator**	Dr. Mohammed Al-Zobbi
	Lecturer & Tutor	Dr. Mohammed AL-Zobbi (m.al-zobbi@aih.edu.au) SYD Dr. MD Ezazul Islam (m.islam@aih.edu.au) MEL
Resources	**Prescribed Textbooks**	Santos, O 2018, Developing Cybersecurity Programs and Policies, 3rd Edn. Pearson.
	Reference Textbooks	Refsdal, A, Solhaug, B and Stolen, K 2015, Cyber-Risk Management, SpringerBriefs in Computer Science. Hubbard, DW, and Seiersen, R 2016, How to measure anything in cybersecurity, John Wiley & Sons.
	Reference Journals	Bentley, M, Stephenson, A, Toscas, P and Zhu, Z 2020, 'A multivariate model to quantify and mitigate cybersecurity risk', Risks, vol. 8, no. 2, p.61. Eling, M 2020, 'Cyber risk research in business and actuarial science', European Actuarial Journal, pp.1-31. Habibzadeh, H, Nussbaum, BH, Anjomshoa, F, Kantarci, B and Soyata, T 2019, 'A survey on cybersecurity, data privacy, and policy issues in cyber-physical system deployments in smart cities', Sustainable Cities and Society, vol. 50, p.101660. Hutchins, MJ, Bhinge, R, Micali, MK, Robinson, SL, Sutherland, JW and Dornfeld, D 2015, 'Framework for identifying cybersecurity risks in manufacturing', Procedia Manufacturing, vol. 1, pp.47-63. Pandey, S, Singh, RK, Gunasekaran, A and Kaushik, A 2020, 'Cyber security risks in globalized supply chains: Conceptual framework', Journal of Global Operations and Strategic Sourcing. Sawik, T 2021, 'Balancing cybersecurity in a supply chain under direct and indirect cyber risks', International Journal of Production Research, pp.1-17.
	Recommended websites:	Australian Cyber Security Centre. Applying a risk-based approach to cyber security. Available from: https://www.cyber.gov.au/acsc/view-all-content/guidance/applying-risk-based-approach-cyber-security [Accessed: 12 May 2021] Britt, P 2017, 'Cybersecurity Risk Management: Finding and Fixing Your Security Vulnerabilities', Available from: https://www.esecurityplanet.com/networks/cybersecurity-risk-management-finding-and-fixing-your-security-vulnerabilities/ [Accessed: 12 May 2021] The European watch on cybersecurity & privacy. Demystifying the cyber risk management process. Available from: https://cyberwatching.eu/demystifying-cyber-risk-management-process/ [Accessed: 12 May 2021]

The AIH Referencing Guide is available to download. Go to the Library and Learning Support Moodle page.

Weekly Workshop Schedule

Week	Week Starting	Session Reading Materials	Workshop Activities	Assessment Task Due
1	Session 1	Session 1 – Introduction: Information Security, Cyber Security & Risk Management	Discussion Forum	
	Session 2	Session 2 – Cyber Systems and Cybersecurity Framework	Discussion Forum	A1: Portfolio activities (Individual) (10%)
2	Session 3	Session 3 – Cyber-Risk Management	Discussion Forum	
	Session 4	Session 4 – Risk Identification	Discussion Forum	A1: Portfolio activities (Individual) (10%)
3	Session 5	Session 5 – Risk Analysis	Discussion Forum	
	Session 6	Session 6 - Quantitative Risk Analysis and Qualitative Risk Analysis	Discussion Forum & Assessment 2	A1: Portfolio activities (Individual) (10%) A2: Quiz (Individual) (30%)
4	Session 7	Session 7 - Risk Evaluation and Risk Treatment		
	Session 8	Session 8 – Risk Measurement	Assessment 3	A3: Group Report & Presentation+individual viva (40%)

Assessment Details

Assessment Number & Type	Assessment Description	Week/ Session Assessed & Due Date	Weighting	Learning Outcomes Assessed	Submission Requirements
A1 Participation Individual	Students will build up a portfolio based on their weekly understanding of concepts. Students will read application case study from their prescribed book/e-text and write a short recommendation for decision makers (150 words).	Session 1-6	30%	ULO1 ULO2 ULO3	In Class 30 minutes
A2 Quiz Individual	Students will be assessed to demonstrate their understanding of the content covered in sessions 1-5.	Session 6	30%	ULO1	Moodle 30 minutes
A3 Report Individual	Students will work on a cyber security risk management project report (25%) and presentation (5%) in a group of 4-5 members along with individual Viva (10%).	Session 8	40%	ULO1 ULO2 ULO3 ULO4 ULO5	Moodle 1 hour
Participation					

NOTE: Pass requirements

Students must make a reasonable attempt in their assessment events in order to be eligible to pass the subject and must achieve a combined result from these assessment events of at least 50% to pass the subject.

If a student fails any individual piece of assessment they will be eligible to continue in the subject and attain a pass in the subject, provided their combined result from all assessment events in the subject is 50% or above.

You can view details on your assessments at Moodle in the Assessment Brief and in-class instruction.

Overall Grade Assessment

Grade	Marking Criteria/Standards
High Distinction (outstanding performance) Code: HD Mark range: 85% and above	Complete and comprehensive understanding of the unit content; development of relevant skills to an outstanding level; demonstration of an extremely high level of interpretive and analytical ability and intellectual initiative; and excellent achievement of all major and minor objectives of the unit.
Distinction (very high level of performance) Code: D Mark range: 75-84%	Very high level of understanding of the unit content; development of relevant skills to a very high level; demonstration of a very high level of interpretive and analytical ability and intellectual initiative; and comprehensive achievement of all major and minor objectives of the unit.
Credit (high level of performance) Code: C Mark range: 65-74%	High level of understanding of the unit content; development of relevant skills to a high level; demonstration of a high level of interpretive and analytical ability and achievement of all major objectives of the unit; some minor objectives not fully achieved.
Pass (competent level of performance) Code: P Mark range: 50-64%	Adequate understanding of most of the basic unit content; development of relevant skills to a satisfactory level; adequate interpretive and analytical ability and achievement of all major objectives of the unit; some minor objectives not achieved.
Non-graded Pass Code: NGP	Successful completion of a unit assessed on a pass/fail basis, indicating satisfactory understanding of unit content; satisfactory development of relevant skills; satisfactory interpretive and analytical ability and achievement in all major objectives of the unit.
Fail - unsatisfactory performance Code: F Mark range: below 50%	Inadequate understanding of the basic unit content; failure to develop relevant skills; insufficient evidence of interpretive and analytical ability; and failure to achieve some or all major and minor objectives of the unit.
Fail - no assessment submitted Code: FNS	Did not present any work for assessment, to be considered as failed.
Grade Pending Code: GP	A final grade is yet to be awarded for the unit as students have not yet completed all the assessment tasks and special consideration has been approved due to medical grounds or exceptional circumstances beyond the control. This is a temporary grade only and must be finalised before the end of the following semester.
Supplementary Exam Code: SX	A final grade is yet to be awarded for the unit as a supplementary exam has been approved for students who have failed the unit and obtained total mark between (46-49%), provided the student has completed all other assessment units. This is a temporary grade only and must be finalised before the end of the following semester.
Withdraw with Failure Code: WF	Cancelled enrolment in the unit after the final date for withdrawal without failure. Cancelled enrolment in the unit due to non-payment of fees.
Withdraw Without Failure Code: AW	Cancelled enrolment in the unit before the final date for withdrawal without failure. This grade may also be awarded to students who withdraw from a unit after the withdrawal date under special or compassionate circumstances. In these cases, the grade is awarded at the discretion of the Teaching and Learning Committee. A unit with the grade of AW does not appear on a student's academic transcript.
Advanced Standing Code: AS	Credit has been granted for the unit of study following an application and its approval for Advanced Standing

Policies & Forms

All students should be familiar with the relevant policies and how they apply to their study. Policies and forms can be accessed on the Australian Institute of Higher Education website at: www.aih.nsw.edu.au

ACCESS AND EQUITY

The Australian Institute of Higher Education is committed to providing equal opportunity and promoting inclusive practices and processes for all students and clients within the limits of its resources.

SPECIAL CONSIDERATION

You may seek Special Consideration in the evaluation of your subject assessments where you feel such matters as serious illness, or other exceptional circumstances beyond your control that may have affected your performance. A Special Consideration form must be submitted within 3 days of the assessment due date or event. Please contact Student Services (studentservices@aih.nsw.edu.au) for a copy of the application form (https://aih.nsw.edu.au/about-us/policies-procedures/).

ATTENDANCE POLICY

The Australian Institute of Higher Education emphasizes the need for all students to attend classes on a regular and consistent basis to develop the skills and attitudes necessary to compete in the highly competitive labor market. Students who are not in attendance for at least 50 percent of any scheduled class will be considered absent for that class. Lateness is a disruption of a good learning environment and is discouraged.

ACADEMIC INTEGRITY

The Australian Institute of Higher Education considers academic honesty to be one of its highest values. Students are expected to be the sole authors of their work (except in the case of group assignments). Use of another person's work or ideas must be accompanied by specific citations and references. The Australian Institute of Higher Education subscribes to a third-party plagiarism detection service, and reserves the right to check all student work to verify that it meets the guidelines of this policy. Academic dishonesty is a serious offense and may result in a range of penalties. Procedures for processing plagiarism, contract cheating and exam cheating offenses are described in the Academic Misconduct Policy and Procedure (https://aih.nsw.edu.au/about-us/policies-procedures/).

ACADEMIC APPEALS

The Student Grievance Handling and Resolution Policy and Procedure policy outlines the process for a fair, equitable and confidential means of resolving complaints and appealing academic decisions (https://aih.nsw.edu.au/about-us/policies-procedures/).

REVIEW OF ASSESSMENT

Where a student believes their assessment instrument has been marked incorrectly, they may apply to have that assessment instrument re-marked once only. Requests must be made within five (5) days of results being released. The Student Assessment Policy and Procedure provides instructions on requesting re-marks (https://aih.nsw.edu.au/about-us/policies-procedures/).

LATE POLICY

An assessment item submitted after the assessment due date, without an approved extension or without approved mitigating circumstance, will suffer a late submission penalty. The late submission penalty is a reduction of the mark allocated for the assessment item by 5% per day (or part thereof) of the total marks available for the assessment item as indicated in the Unit Outline. **A "day" for this purpose is defined as any day of the week including weekends**. Assignments submitted later than one (1) week after the due date will not be accepted (https://aih.nsw.edu.au/about-us/policies-procedures/).

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